

Project Scope: **lecture 4**

Maximum Grade: 5

Student Name:

Student ID:

Assignment Rules & Regulations:

- Plagiarism, cheating, or any form of academic dishonesty is **strictly prohibited**. All work submitted must be the student's own, and all external sources, such as code snippets, libraries, or academic papers, must be properly cited.
- Requests for extensions must be requested **in advance of the deadline** and should be supported by valid and documented reasons, such as medical or family emergencies. Extensions will be granted at the instructor's discretion.
- If you encounter any issues or require clarifications related to the assignment, promptly communicate with the instructor or teaching assistant. **Do not share your assignment solutions with classmates before the submission deadline.**

Mini-Project Objective:

- Curve fitting.

It is required to fit the data given in the file entitled "data_to_curve_fit" with an appropriate closed-form function, so as to minimize the three key performance indicators (KPIs), namely, the 2-norm of residuals, the squared error (SE), and the root mean squared error (RMSE) (recall lecture 4, slide 40). The fitting function may be piecewise defined.

Deliverables (individual submission):

1. The code for data importing, conversion, plotting, curve fitting through appropriate models, and the evaluation of different KPIs.
2. The graph of the fitting function on the same plot of the given data points.
3. A technical report summarizing the code, fitting function, graphs, different KPIs.
4. A soft copy and a hard copy of each of the above requirements have to be submitted.

Deadline for Submission and Discussion: Sunday, December 21st, 2025.

